

Seminar Presentation

Contrastive Learning for Sequential Recommendation

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Xu Xie et al. (2022). "Contrastive learning for sequential recommendation". In: *2022 IEEE 38th international conference on data engineering (ICDE)*. IEEE, pp. 1259–1273

Introduction – Sequential Recommendation

- ▶ **Recommender Systems**

- Help users find relevant items (e.g., products, movies, news).
- Many approaches assume static user preferences.

- ▶ **Sequential Recommendation**

- Recommendations based on a user's interaction history.
- Able to better capture dynamic user interests over time.

Introduction – Sequential Recommendation

- ▶ Formulation as a prediction task
 - Given a historical interaction sequence of a user.
 - Predict the next item the user will interact with.
 - Very similar to the language modelling task in NLP.

Introduction – Challenges in Sequential Recommendation

- ▶ Data Sparsity

- Most users and items have only a few interactions.
- Difficult to learn robust user preference patterns from short sequences.

- ▶ Learning Effective User Representations

- Deep models have many parameters, need lots of data.
- Next-item prediction alone can overfit and struggle to generalize.

Introduction – Contrastive Learning

- ▶ Self-Supervised Learning

- Learning from the data itself without human-created labels.
- Define some pretext task based on the raw data.
- Has been effective in other domain like CV and NLP.

- ▶ Contrastive Learning

- The pretext task is to compare different instances.
- The model learns to tell similar things apart from dissimilar things.
- Positive pair: Two augmented versions from same input.
- Negative pair: Augmented versions of different inputs.

CL4SRec – The Proposed Framework

- ▶ User Representation Encoder

- Takes a user's interaction sequence as input.
- Produce a user representation as output.
- Can be implemented using different models.

- ▶ Encoder Shared Between Two Learning Objectives

- Recommendation Loss: Standard task of predicting next item in the sequence.
- Contrastive Loss: Compare augmented views from the original interaction sequences.

CL4SRec – Data Augmentation Strategies

- ▶ Item Crop → Randomly select a continuous subsequence of the sequence.
- ▶ Item Mask → Randomly replace some items with a special mask item.
- ▶ Item Reorder → Randomly shuffle items within a continuous subsequence of the sequence.

CL4SRec – Model Architecture

- ▶ User Representation Model based on Transformer
 - Items and position embedding layer.
 - Multi-head self-attention
 - Position-wise feed-forward network
 - Layer normalization and dropout for regularization
- ▶ Next item probability prediction
 - Based on dot product between user and item representation.
 - The items closest to the user representation are predicted.

CL4SRec – Training Process

Evaluation – Experimental Setup

Evaluation – Comparison With Other Methods

Evaluation – Influence of Hyper-Parameters

Evaluation – Data Augmentation vs. Contrastive Learning

Evaluation – Better User Representations

Evaluation – Strengths and Weaknesses of the Paper

Quiz

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